“Mapping the Vanishing and Reappearing Quanta Points of the IVM’s 6 Axis Metric Upon the Cartesian Coordinate System’s 3 x,y,z axis”

Abstract:

In Part 1 I will explain how Synergetics geometry Abrogates, or makes defunct, Goedel’s Incompleteness Theorem.

In Part 2 of this paper I will list the x,y,z coordinates for the placement of the closest packed unit radius spheres which constitute various polyhedron’s sizes and shapes within the Isotropic Vector Matrix or IVM. I wrote this paper because Synergetics students have asked if I could help them to visualize my various vector constructed polyhedrons by displaying them on a computer screen as rotatable 3 dimensional images. Since I have no computer programming skills I’ve listed these IVM coordinates as x,y,z coordinates. This will enable computer programmers to input the correct x,y,z coordinates which will enable them to show the various polyhedrons and their transformations as rotatable vectors as well as showing how the spheres transform into spaces and vice versa via a rotatable 3 dimensional visual display.

I have listed the coordinates in 2 separate ways. In the first I have oriented the tetrahedron so 2 of its oppositely oriented edge lengths are running along the x and y axis of x,y,z. This shows the tetrahedron’s edge length increasing by 1 unit vector, or 1 inch increments. This is NOT the way the tetrahedron is oriented within the x,y,z system but it is the easiest way to calculate the coordinated points and it also shows the incremental edge length increase in units of 1 inch vectors. With this orientation we can ‘see’ the edge lengths increasing in whole rational units of a 1 inch unit vector.

The second set of coordinates are what might be thought of as the ‘Correct’ set because this is exactly how the polyhedrons of the 6 axis metric are oriented within the 3 x,y,z axis metric and should be thought of as the preferred way to orient the IVM.

Introduction:

If we plot out all of the following x,y,z coordinates into a 3 dimensional computer display and use each coordinate as the placement for the center of volume of a sphere with a 1 inch diameter we can construct a visual representation of the spheres which closest pack into an omni-concentric hierarchy of different sizes and shapes of 6 symmetrical polyhedrons.

If we can show each 1 inch diameter sphere shrinking to an infinitesimally small size we will be able to map out the touching points between the shrinking spheres as lengthening vector lines. This will enable us to visualize the structures in their sphere state or their vector state. When the spheres shrink to ‘zero size’ the touching points between the spheres become vectors which will have extended to 1 inch. The vectors will now all touch at the ‘dimension setting’ point at the

center of the sphere. The reduced spheres represent touching ‘points’ where all the vectors interconnect. The reduced vectors represent touching points where all the spheres interconnect.

When the vectors reduce their size to ‘zero points’ and ‘vanish’, the spheres appear, and when the spheres reduce their size to zero points and vanish, the vectors appear.

The vectors interconnect the center of volume of all the adjacent spheres and therefore are the same size as the diameter of the sphere, 1 inch. In this way one can shrink the 1 inch diameter sphere to a ‘point’ and see the 1 inch ‘vector lines’ lengthening to interconnect the ‘dimension setting points’ to reveal the different sized and shaped polyhedrons within the matrix.

These points and lines reveal an unlimited number of frequency related ‘size assemblages’ of variously ‘shaped’ polyhedrons within the isotropic vector matrix and this helps us to understand what fuller was talking about in 982.61 about an omniconcentric hierarchy of rationally phasing polyhedrons. With this model we can see how the isotropic vector matrix’s tetrahedral frame creates a series of polyhedrons which are all phasing into and out of the center of volume of the tetrahedron's center to create the omniconcentric hierarchy of polyhedrons. We can ‘see’ tetrahedrons, octahedrons, truncated tetrahedrons, cubes, cuboctahedrons and tetrakaidecahedrons all developing within one another and within the internal center of volume of the original expanding tetrahedral frame starting from one central sphere point or ‘singularity’ and expanding omni-directionally from this one starting point.

If the programmer enables the ‘growth’ of the polyhedrons to be shown as a transformational process, it should be possible to show each of the spheres from any polyhedron to shrink into the infinitesimally small point at the center of the 3 spaces within the metric.

As an example: We may show the first 1 inch diameter white sphere of the zero frequency tetrahedron shrinking to a white point as the four red spheres of the 1 frequency tetrahedron begin to grow from 4 infinitesimally small red ‘points’ into four 1 inch diameter spheres.

The first white sphere disappears and becomes the center of volume of the 1 frequency tetrahedron’s 4 red spheres.

Each next size, or frequency, of tetrahedron will have all the spheres from the previous frequency shrink and transform themselves into the center of volume of all the tetrahedral spaces of the next frequency tetrahedron.

These 4 red spheres from the 1 frequency tetrahedron will then reduce to red points within the center of volume of the 4 new tetrahedral spaces which ‘appear’ when the 2 frequency tetrahedron’s 10 spheres ‘phase’ into their ‘coordinated spaces’.

The 3 separate spaces within the matrix will have these ‘points’ at the center. The point becomes revealed at the center of the spaces as all the spaces within the complex ‘transform’ themselves one into another.

As an example the white sphere at the center of the complex transforms into a tetrahedral space at the center of the 1 frequency tetrahedron, then into an octahedral space at the center of the 2 frequency tetrahedron, and then into an inverted tetrahedral space at the center of the 3 frequency tetrahedron, before transforming back into the white sphere at the center of the 4 frequency tetrahedron.

This omni-directionally expanding tetrahedral frame of reference shows the spheres transforming themselves into the 3 separate spaces, tetrahedral, octahedral and inverted tetrahedral spaces, before re-transforming into a ‘point particle sphere’ every fourth frequency. This omni-directionally expanding tetrahedral frame’s center of volume then reveals an omni-concentric hierarchy of 6 rationally phasing, internally developing, symmetrical polyhedrons as ‘appearing, vanishing and reappearing’ quanta points.

It is in this way that students may gain a better visualization of the ‘process’ which the particle goes through as it transforms itself from a sphere into a ‘wave pattern integrity’ and then back into a point particle sphere again.

If programmed properly the center white sphere never moves from (0,0,0), it simply ‘vanishes’ and then transforms itself at the center of the matrix into the 3 separate spaces before it reappears as a 1 diameter sphere at the center of the 4 frequency tetrahedron which holds the 1 frequency cuboctahedron or vector equilibrium at its center of volume. The central sphere will then vanish into the central tetrahedral space of the 5 frequency tetrahedron and then retransform itself into a sphere at the center of the complex at the 8th frequency and this central sphere will then reappear at every 4th frequency of the tetrahedron’s development.

We will notice that none of the spheres from any frequency polyhedron are at the same coordinates for that same polyhedron’s next frequency’s coordinates.

Part 1: Synergetics Abrogates Goedel’s Incompleteness Theorem.

This visualization process will help us to understand what Arthur Eddington was talking about when he explains that the subatomic particles transforming into one another constitute a ‘reshuffling of the points of the complex”. We can ‘see’ how the ‘reshuffling of the points of the complex’ enables the creation of many different sizes and shapes of various inter-transforming ‘point particle polyhedrons’ all of which have exact numbers of tensor points, vector edge lengths, and tetrahedral volumes. This should enable future researchers to discover ratios within the IVM which can be ‘coordinated’ with the exact transformational ratios between all of the exotic short lived subatomic particles.

We may recall from my previous papers that Universe can ‘reshuffle’ the 1 central ‘electron point’ into and out of the center of the 17 frequency vector equilibrium to transform its 1838.5 neutron ratio into its 1836 proton ratio which gives us a geometrical model as to how and why

the weak force interaction occurs when neutrons decay into protons and 1 electron leaves the neutron system to become a proton system.
(See video here: https://www.youtube.com/watch?v=E_4JucYbkEY)

Although my work has correlated many aspects of nuclear structure it has not identified these exotic subatomic particles but Jeff Yee's work has. This leaves me to hypothesize that we should be able to 'account' for these same particles within the IVM with further research by examining the vector totals within specific frequencies of the various polyhedrons within the IVM. See Jeff's paper: "The Geometry of Particles and the Explanation of their Creation and Decay". Available here: https://www.rxiv.org/pdf/1908.0193v1.pdf

As Arthur Young explains in "The Geometry of Meaning", "It has been discovered that nuclear particles are permutations, reshufflings, of several quite exoteric factors or parameters. Some of the parameters are little more than symbolic: spin, strangeness, charge, atomic mass. But these permutations or reshufflings produce a number of distinct entities: pi and kappa mesons, lambda, sigma and xi hyperons etc, all of which are transitory; ie, they disintegrate almost immediately. Only two permutations are permanent: the proton and the electron."

Arthur explains, "When nuclear particles radiate, or rather, dissipate into radiation (for they do not remain particles) the dissipation products are other particles as well as radiation, adding thus an extra "dimension".

We may note that this extra dimension is related to the 'phase space' of Eddington's 'fifth phase dimension'. This is how and where all the different mathematical and philosophical ideas for "extra dimensions" began. We can now see how this 'phase space' relates to the visually apprehensible model of the point particle vanishing and transforming itself through the 3 'phase spaces' before reintegrating itself into a point particle at the center of the complex every fourth frequency.

Young quotes Eddington, "There is a certain hiatus in the arguments for the relativity theory which has never been thoroughly explored. We refer all phenomenon to a system of coordinates, but do not explain how a system of coordinates (a method for numbering events for identification) is to be found in the first instance. It may be asked, What does it matter how it is found, since the coordinate system is entirely arbitrary in the relativity theory? But the arbitrariness of the system is limited. We may apply any continuous transformation; but our theory does not contemplate a discontinuous transformation of coordinates such as would correspond to a reshuffling of the points of the continuum."

What he is talking about is related to Goedel's incompleteness theorem which basically states that any system of geometry can be created upon any set of axioms and since all such geometries are based on their own self asserted axioms they cannot be considered as a 'complete' system because they thereby, inherently by design, do not include all other potential geometries which can be generated, each with their own underlying axioms.

This meant that there could be no “Theory of Everything” because no all-encompassing mathematical ‘theory’ could prove all mathematical truths; but the Symmetrical Polynomial Equations from Synergetics demonstrates that it is consistent because it proves its own mathematical truths by using the underlying omni-symmetry of the Tetrahedron’s internally structured Isotropic Vector Matrix as Nature’s coordinate system for referencing all kinematic energy momentum characteristics of Universe’s intertransformative polyhedral systems in whole rational units of energy vectors, gravity tensors and tetrahedral volumes. This system is free from internal contradictions and is consistent with uniting various other systems of math and geometry as I shall herein extrapolate.

Goedel’s Theorem is related to what Crowley was talking about in his paper “What is a Number or a Symbol?” where he states, “Bolyai, Lobatschewsky and Riemann have shown conclusively that a consistent system of geometry can be erected on any arbitrary axiom soever. If one chooses to assume that the sum of the interior angles of a triangle is either greater than, or less than, two right angles, instead of equal to them, we can construct two new systems of geometry, each perfectly consistent with itself, and we possess no means soever for deciding which of the three represents truth”.
(Liber 777. Page 134: “What is a ‘Number’ or a ‘Symbol’?” by Aleister Crowley
http://93beast.fea.st.user.fm/files/section1/777/Liber%20777%20Revised.pdf)

This is directly related to what Stephen Hawking is explaining in his article “The (Elusive) Theory of Everything” in the August 2013 edition of ‘Scientific American’ where he explains ‘why’ all of our attempts to develop a unified field theory, or a theory of everything will continue to ‘elude’ us.

First Stephen explains that if we were a fish living in a round fishbowl we would perceive things moving outside of our fishbowl as always moving in ‘curved lines’ even though the people walking outside the bowl would perceive that they are walking in ‘straight lines’.

He explains that, if we wanted to, we could create a mathematical and geometrical system which could ‘prove’ that the people outside the fishbowl really were all walking in ‘curved lines’.

Hawking uses this fact of mathematical abstraction as an analogy to justify why he thinks we will never understand the unified field, as he explains how there are 5 different versions of string theory, each of which describes its own version of reality and this is why he thinks, “It might be that to describe the Universe we have to employ different theories in different situations”.
https://www.scientificamerican.com/article/the-elusive-theory-of-everything-extreme-physics-special/

This helps us to understand how Goedel’s ‘Incompleteness Theory’ acts as an ‘impediment’ in our attempt to understand Universe and unify the four fundamental forces with the masses of the subatomic particles. (Pg, 109-110, ‘The Geometry of Meaning’)

Goedel’s Theorem is just one of several different theorems which ‘enables’ mathematical physicists to ‘believe’ that we will never be able to ‘discover the Unified field’. Heisenberg’s

Uncertainty ‘Principle’ is another one of these theorems which Academia ‘believes’ in and so this ‘theory’ and others continue to act as impediments to our comprehensive understanding of the complex considerations involved because it prevents humans from investigating areas of research which they think they “already know will not lead to fruition”.

Our belief systems keep our minds locked into a specific reality and this hampers us from investigating outside of those belief systems already preset parameters. A very comprehensive paper was written by Andre Michaud which explains in detail how the uncertainty principle developed and why it is not a principle or law of physics, but only a well accepted Axiom. See his paper “Critical Analysis of the Origins of Heisenberg's Uncertainty Principle here: https://www.gsjournal.net/Science-Journals/Research%20Papers-Quantum%20Theory%20/%20Particle%20Physics/Download/9885

Synergetics abrogates Goedel’s incompleteness theorem because it is not based on mathematical or geometrical axioms. It always identifies all aspects of real physical Universal structure in relation to the underlying 6 axis metric of the IVM which is disclosed by correlating pre-time, pre-size closest packed unit radius spheres with their frequency coordinated polyhedral continuums which enable whole rational unit ‘accounting’ of ‘dimension setting’ energy vectors, gravity tensors and tetrahedral volumes in relation to all special case event scenarios in Universe. “Time and size and special case physical reality begin with frequency”. (Section 1033.601)

This is why I’ve been able to correlate different geometrical and mathematical models related to nuclear structure with the IVM of synergetics geometry.

We will recall that I correlated the 6 dimensions within string theories Calabai Yau Manifold and its 6 positive and 6 negative degrees of freedom related to the ‘tangent bundle point’ upon the manifold with synergetics IVM and the VE as the intertransformative ‘point’ upon the complex with its internally structured 6 axis metric. I correlated the equations which set the “Symmetry of Dimensions” within their SU(n) and U(n) Groups as n^2-1 and n*1 with the total number of ‘symmetrical dimension setting’ vectors within the tetrahedron’s structure which is equal to (n^2-1) * (n*1) = n^3-n when n = the total number of spheres on the tetrahedron’s edge length. I also correlated their equation x^2=3y, which is related to their ‘Equality Case’, which states that “the ‘square’ of the first Chern Class is equal to 3 times the second Chern Class”, with the total number of electrons on the s,p,d and f subshells of atomic structure using synergetics cube’s volume ratios as I explain in my video.
(See Video Here: https://www.youtube.com/watch?v=JzYjLR1oXEo)
[See equations for string theory on pages 203 and 122 in Shing Tung Yau’s book “The Shape of Inner Space” available here: https://zr9558.com/wp-content/uploads/2014/11/shing-tung_yau_nadis_s-_the_shape_of_inner_space.pdf]

We will recall that I’ve also shown how synergetics symmetrical equations take the place of the Hamiltonian Quaternions and complex numbers and how this correlates nicely with Andre

Michaud's equations for the electromagnetic mechanics of subatomic particles and his 3 spaces model which uses the Hamiltonian Quaternions and the complex numbers to transition into the x,y,z coordinate system where i, j and k represent momentum vectors in 3D space.

As can be seen in this paper I am also correlating all the energetic vectorial 'lines' and gravity tensor 'sphere points' from synergetic's 6 axis metric with the 3 x,y,z axis coordinate system as pre-time, pre-size, pre-existing ratios.

In 964.31 Fuller explains that "In the quantum and wave phenomenon, we deal with individual packages. We do not have continuous surfaces (or transformations). In synergetics we find the familiar pattern of second powering (c^2 or F^2) displaying a congruence with the points, or separate little energy packages, of the shell arrays. Little energy actions, little separate stars: this is what we mean by quantum. Synergetics provides geometrical conceptuality in respect of energy quanta."

We see that synergetics does contemplate a 'discontinuous transformation of coordinates' which corresponds to the 'reshuffling of the points of the continuum' as discontinuous little energy package points, omnidirectionally expanding from one central 'sphere point' as the growth of the external shell layer of closest packed unit radius spheres within the tetrahedron's structure at the rate of 2F^2+2. The first 6 frequencies of these omni-directionally expanding tetrahedral 'points' from Nature's 6 axis coordinate system are herein being correlated with mankind's 3 axis system as (x,y,z) coordinates.

We can visualize that the 'discrete' discontinuous points upon the surface and within the internal structure of the tetrahedron must transition through 3 separate discrete and discontinuous 'phase spaces' during their transformation from one discrete 'point in space and time' to the next. This is related to Andre Michaud's 3 spaces model and helps us to understand the process the particle goes through during its phase transition process through the IVM.

Eddington states, "The fundamental basis for all things must presumably have structure and substance". Young connects substance and structure as he asserts "substance correlates with the 'ordering' of the points of the continuum". "The difference between the atomic and nuclear realms is of a rather abstract nature".

Young explains, "Our interpretation holds that the originating and prime 'stuff' is unitary, undifferentiated, devoid of properties, without form or substance; it is known to physicists as radiation". (Pg. 88, "The Geometry of Meaning")

So Arthur thinks that the prime originating stuff that all the subatomic particles are made of is electromagnetic radiation, EMR.

The atomic realm is governed by well known principles of physical structure related to molecular physics and chemistry but the nuclear realm is not so well understood.

Arthur explains that “recent developments in nuclear physics” discovered that “the several different types of particles are ‘accounted’ for as permutations of some more basic entities” and this confirms the “insights of Eddington some twenty five years before the complexity of the subnuclear realm was known”.

He explains, “We must realize that ‘points’ are undefined entities. And so are quarks. We cannot endow these elements themselves with spatial or even substantial nature, because they are only carriers of abstract properties. It is the ‘ordering’ of quarks which produces what we recognize as mass or charge”.

We can see how the ‘discontinuously-phased ordering’ of the points on the continuum produces the different sizes and shapes of all the polygons and polyhedrons.

Young quotes Gamow, “Although experimental studies of these new particles reveal new and exciting facts about them almost every month, theoretical progress in understanding their properties is almost at a standstill. We do not know why they have the masses they do. We do not know why they transform into one another the way they do. We do not know anything”.

Although Gamow was writing in 1959, Academia scientists still have no answers for these and other questions related to subnuclear physics.

As we may recall, I’ve explained how, at specific frequencies of the tetrahedrons structure, the vector sum ratios correlate the Mev/c^2 of the electrons, protons, neutrons and quarks with Planck’s constant, the fine structure constant and the gravitational constant in whole rational units using only the basic math and geometry of the symmetrical polyhedrons from synergetics and their associated symmetrical polynomial equations.

It seems that the underlying ‘substance’ for the subatomic particles is electromagnetic radiation and the process Universe uses to structure this primordial and primitive ‘substance’ may be related to the symmetry of the underlying expanding and contracting tetrahedral field which acts as a model for all the unlimited frequency ranges of EMR. As can be seen in the picture of the 6 frequency tetrahedron there are 6 different sizes of tetrahedron all bundled up inside of the larger tetrahedron and they are all ‘Holographic Images’ of one another.

The underlying symmetry of our holographic Universe seems to be based on the tetrahedon’s ability to continuously fractionate face planes or continuously add face planes and never lose the symmetry of the underlying intertransformative tetrahedron’s structural frame of reference. No other symmetrical polyhedron can have a face plane ‘sliced’ off of it or ‘added’ to it and still be perfectly symmetrical with all the faces and edge lengths still maintaining the same size to one another. The tetrahedron’s 4 faces will all still be of equal area and the 6 edge lengths will all be equal length. (See sections. 623.10, 986.21, 986.061-986.064)

Since the Hamiltonian Quaternions are used by Andre Michaud as a mathematical and geometrical correlation with his 3 spaces model to demonstrate the processes which EMR uses

to ‘form’ atomic structure, we can draw a clear correlation between the EMR field and the ‘holographic fractal tetrahedral field’ because the symmetrical polynomial equations which calculate the vector, tensor and volume totals of the symmetrical polyhedrons from Synergetics all hold identical algebraic expressions to the Hamiltonian vector and scalar equations, and at specific frequencies the vector sum ratios disclose identical mathematical ratios to the electromagnetic mechanics of subatomic particles. (See Andre Michaud’s Index: https://www.gsjournal.net/Science-Journals/Research%20Papers-Unification%20Theories/Download/2460)

As further proof that electromagnetic radiation is the pure primitive substance from which all particles originate see Yee and Gardi’s papers “Particles of the Universe meets the Electric Universe”, “The Relationship of the Mole and Charge” and “The Relationship of Mass and Charge”. (Available Here: https://www.researchgate.net/publication/330737944_Particles_of_the_Universe_Meets_the_Electric_Universe
Here: https://vixra.org/pdf/1904.0445v1.pdf
Here: https://vixra.org/pdf/1903.0550v2.pdf

In “The Origin of Gravitation and Electromagnetism” Zhong-Cheng Liang confirms Jeff Yee’s work as he shows the relationship between the electron’s mass and charge. As he explains on page 86, “This article will prove that electric charge is an attribute equivalent to the mass of particles”. See his equations (94-98). On page 101 he states, “Eq. (95) clearly shows that the electric charge is equivalent to the mass, and the electric current is equivalent to the momentum” and on page 100 he states, “It is clear that both electromagnetic field and gravitational field originate from the interaction of particles”. (Available here: http://www.isaacpub.org/images/PaperPDF/TP_100085_2019032816260004894.pdf

This is exactly what Michaud is explaining in chapters 3.2 and 3.3 of his “Electromagnetic and Kinematic Mechanics Synchronized in their Common Vector Field” where he explains the origins of the Coulomb interaction as the resultant of the massless and chargeless 1.022 Mev/c^2 photon’s decoupling into the mass/charge of the electron and positron of .511 Mev/c^2 each, thereby creating the potential for the first attractive and repulsive proclivities between particles.

Hopefully the continuing progress from independent explorers will soon be acknowledged by Academia as we continue to pave the way for a clearer understanding as to how and why all of the forces and masses are interoperatively associating and transforming within the nucleus of the atom the way they do.

(See pages 68-72: 88: 109-110 in “The Geometry of Meaning” for more information about Arthur Young’s ideas about Goedel’s, Eddington’s and Gamow’s considerations. Available here: https://archive.org/details/geometryofmeanin00arth

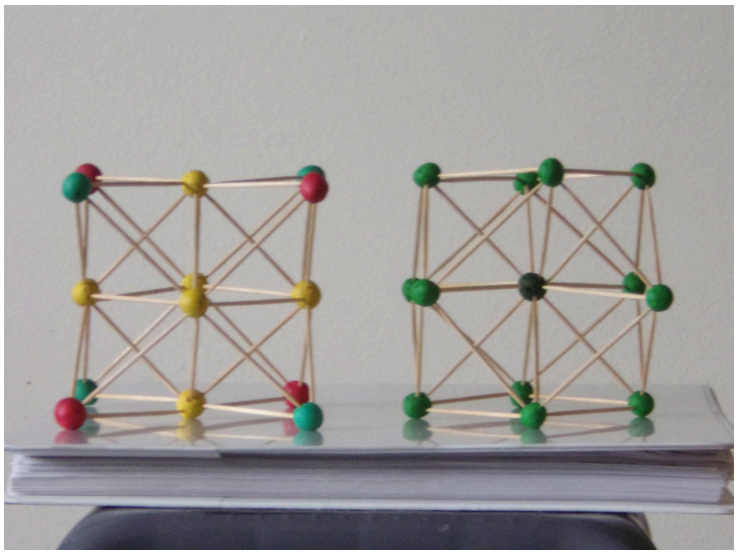
Part 2) Plotting the x,y,z coordinates for the tetrahedron's IVM oriented closest packed spheres.

A) Introduction:

As we place these points be aware that the tetrahedron is oriented so it is 'sitting up' on one edge length as can be seen in Figure 1 where we have a 1 frequency cube next to a 1 frequency cuboctahedron, or Vector Equilibrium, or VE. The cube and the VE are oriented so they are both sitting upon their square faces. The 2 frequency tetrahedron is sitting up on one edge inside of the 1 frequency cube. The cube is 'structured' by intercombining another 2 frequency tetrahedron which is oppositely oriented.

As can be seen in Figure 1, the cube's edge length is 1.414213562 compared to one toothpick vector. We will give the toothpick vectors a measure of 1 inch. This 2 inch edge length tetrahedron has a 1.414213562 inch distance between its oppositely oriented edge lengths. This means that the 1 inch edge length tetrahedron has a .7071067 distance between its oppositely oriented mid edge length 'points'. One half of .7071067 is the center point of the tetrahedral complex and this is why you will note that the 1 frequency tetrahedron's 4 vertices set the coordinates for the 4 sphere points at .3535533 above and below the (0,0,0) starting point on the xyz axis.

Figure 1: 1 frequency cube and 1 frequency cuboctahedron or VE.



B) 2 Separate Orientations of the Tetrahedron within the x,y,z System.

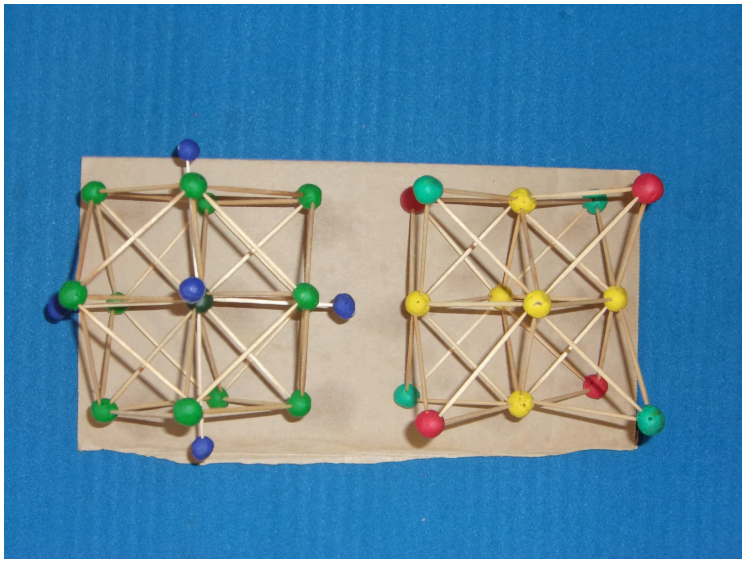
The blue spheres in the picture below represent the 3 x,y,z axis. All sphere coordinates listed in the next section 'C' of this paper are orienting the polyhedrons in this manner. In this section 'B' we will be rotating the complex about the 'z' axis by 45 degrees.

If we were to place the blue spheres and the 3 x,y,z axis into the cube on the right we would notice that the x,y,z axis run through the octahedron's 6 vertices which are shown as 6 yellow clay ball sphere points. This is the 'correct' orientation of the x,y,z axis in relation to the polyhedrons of the IVM as listed in section 'C'.

In this section 'B' we will be orienting the red tetrahedron so its edge length as shown in the cube on the right is the x axis and the blue tetrahedron's edge is the y axis. The z axis runs perpendicular to these and connects the 2 'top' and 'bottom' yellow spheres of the octahedron. Notice that when we add the 3 xyz axis to the IVM's 6 axis it synergetically intercombines with these vectors to create 4 separate sets of 3 axes all of which are orthogonally oriented at 90 degrees to one another.

Figure 2: 1 frequency VE's 6 axis shown with green balls are correctly oriented with the 3 x,y,z axis shown with blue balls. On the right is the 1 frequency cube.

Note that in this first set of coordinates in section 'B' the 2 red sphere points on the face of the cube are the x axis and the 2 blue sphere points are the y axis. If we 'spin' these 2 x and y axis 45 degrees about the z axis they will align themselves with the blue balls as shown in the VE on the right and this is how the second set of 'Correctly Oriented' coordinates are set in section 'C'.



Since every regular tetrahedron's oppositely oriented edge lengths will always run at 90 degrees to one another and they will always be .7071067, or the second root of 1/2, distance between one another compared to its 1 inch edge length, we can orient these two oppositely oriented 1 inch vector edges on the x and y axis as length and width variables and we can use the distance between these oppositely oriented x and y axis vector edge lengths to measure the height of the tetrahedron on the z axis. (Also: See Figure 7 below.)

The reason I'm explaining this is so we can understand the way we are orienting the tetrahedron within the x,y,z, matrix and why we are placing the points where they are.

If the 1 frequency tetrahedron with a one inch edge length has its center of volume at the (0,0,0) point the height and depth on the y axis will be .7071067. So the first points on the y axis will split this distance between them and the center of the edges will be at (0, 0, .35355339) and (0, 0, -.35355339). The edge length is 1 inch so on the x and z axis we plot the distance between the first vertex points at .5 and -.5.

This give us the coordinates:
(.5, 0, .35355339) (-.5, 0, .35355339) (0, .5, -.35355339) (0, -.5, -.35355339)

If we interconnect these 4 vertex points with lines they will all be 1 inch long and enclose .11785 cubic inches of volume or 1 tetrahedral inch of volume. All polyhedrons listed except the cube will have whole rational edge lengths as 1,2,3,4,5,6… etc… and the vector, tensor and tetrahedral volumes will all be whole rational units of measurement.

This enables us to plot one inch diameter spheres upon the x,y,z axis and create the 6 symmetrical polyhedrons within the tetrahedron's IVM domain using only whole rational numbers of 1 inch diameter spheres as points and whole rational numbers of 1 inch vectors as edge lengths to enclose all space in whole rational units of tetrahedral volume.

Exact numbers for approximate values listed as 'z' coordinates in section 'B' are as follows:
The exact value for .35355339 is the second root of 1/8.
The exact value for .7071067 is the second root of 1/2.
The exact value for 1.06066017 is the second root of 9/8.
The exact value for 1.414213562 is the second root of 2.
The exact value for 2.12132034 is the second root of 9/2.

These are also equal to:
The exact value for .35355339 is the second root of 2 x .25
The exact value for .7071067 is the second root of 2 x .50
The exact value for 1.06066017 is the second root of 2 x .75
The exact value for 1.414213562 is the second root of 2 x 1
The exact value for 2.12132034 is the second root of 2 x 1.5

In this first sequence of coordinates these are all 'height' values on the 'z' axis and can be thought of as whole rational 'units' of the size ".353553" because the values are always whole rational unit 'ratios' between points on the z axis. Notice that in the second sequence of 'Correct' coordinates this height value will be the tetrahedron's 'mid-sphere radius' value and this number correlates all 6 of the tetrahedron's mid edge point's distances from the center of volume of the tetrahedron upon the 3 xyz axis so r, radius = .35355339 x 'F' or Frequency = 1,2,3,4,5,6…etc.
.35355339 x 1 = .353553391
.35355339 x 2 = .707106781
.35355339 x 3 = .1.06066017
.35355339 x 4 = 1.41421356
.35355339 x 6 = 2.12132034

The .35355339 x 5 = 1.767766953 is not listed because these are the height values for the 5 frequency tetrahedron and I did not list those points. All continuing height values on the z axis (or 'insphere radius' on x,y,z axis) for the rest of the tetrahedron's frequency related sphere points are at continuing whole rational unit progressions of this .35355339 value. ie. The 7 frequency tetrahedron's radius values will be 7 x .35355339 and the 8 frequency will be at 8 x .35355339 and so on for the rest of the tetrahedron's frequencies.

(Since triangles and tetrahedrons also accumulate at second and third power rates, we use the words second root for square root and third root for cube root to replace Academia's neurolinguistically programmed stranglehold over the way we visualize squares and cubes when thinking about the second and third power algorithmic progressions of growth and decay related to polygonal and polyhedral 'structures'. Since only triangles and triangularly structured polyhedrons are 'structurally stable' and since Fuller was always trying to be semantically precise he tried not to use the terms 'squaring' and 'cubing' for second and third powering. (See Synergetics Section 600: 608.21-608.32 for definition of structure and see 990.01 which shows how triangular and tetrahedral accounting accumulates area and volume in second and third power progressions identical to square and cubic accounting.)

0 Frequency Tetrahedron: 1 White Sphere: (0,0,0)

Notice how this central white sphere at (0,0,0) vanishes and reappears at the center of the 4 frequency tetrahedron. If we were to map out the 6 'points' where this 1 inch diameter sphere touches the xyz axis we would plot the points at (.5, .5, 0) (.5, -.5, 0) (-.5, .5, 0) (-.5, -.5, 0) (0, 0, .7071067) (0, 0, -.7071067) which we will note is identical to the center of volume of the 6 yellow spheres which constitute the 1 Frequency octahedron's first 6 vertices and which are its first six 1 inch diameter spheres. When the 1 frequency octahedron's 6 spheres appear the first sphere from the 0 frequency octahedron vanishes and becomes the center of volume of the newly formed 1 frequency octahedron.

1 Frequency Tetrahedron: 4 Red Spheres:
(.5, 0, .35355339) (-.5, 0, .35355339) (0, .5, -.35355339) (0, -.5, -.35355339)

2 Frequency Tetrahedron: 10 Spheres:

4 Red Spheres at tetrahedron's vertices:
(1, 0, .7071067) (-1, 0, .7071067) (0, 1, -.7071067) (0, -1, -7071067)

6 Yellow Spheres at 1 frequency octahedrons 6 vertices: (.5, .5, 0) (.5, -.5, 0) (-.5, .5, 0) (-.5, -.5, 0) (0, 0, .7071067) (0, 0, -.7071067)

3 Frequency Tetrahedron: 20 Spheres:

4 Red Spheres at tetrahedron's vertices:
(1.5, 0, 1.06066) (-1.5, 0, 1.06066) (0, 1.5, -1.06066) (0, -1.5, 1.06066)

4 Blue Spheres at inverted 1 frequency tetrahedron's vertices at center of complex.
(0, .5, .35355) (0, -.5, .35355) (.5, 0, -.35355) (-.5, 0, -.35355)

If we superimpose these 4 blue points of the inverted 1 frequency tetrahedron over to the 4 red points of the obverted 1 frequency tetrahedron we will create the .5 frequency cube with a .7071067 edge length and a 1 inch diagonal with 3 tetravolume. Notice that the 8 separate 1 inch spheres will not all fit inside of this same cubed space. The 3 volume cube is a mathematical ratio based on the diagonal being the same length as the internally structured tetrahedron. Use 3V^3 to calculate the cubes 'tetrahedral' volume where V=length of the cube's face diagonal. This equation helps us to understand that the cube in synergetics section 982.43 - 982.44 is the 3 tetra-volume cube with a 1 vector or 1 inch face diagonal and the cube in section 982.45 and its associated picture in figure 986.21 with a 2 vector diagonal is the 24 tetra-volume cube.

12 Purple Spheres at vertices to 1 Frequency truncated tetrahedron:
(.5, 0, 1.06066) (-.5, 0, 1.06066) (1, -.5, .35355) (-1, -.5, .35355) (1, .5, .35355) (-1, .5, .35355) (.5, -1, -.35355,)(-.5, -1, -.35355)(.5, 1, .35355)(.-5, 1, .35355) (0, .5, -1.06066) (0, -.5, -1.06066)

4 Frequency Tetrahedron: 35 Spheres:

1 White Sphere at center of complex: (0, 0, 0)

The white sphere which 'vanished' into the center of volume of the 1 frequency tetrahedron now 'reappears' at the center of the 4 frequency tetrahedron. Every 4th frequency of tetrahedral 'phasing' the vanishing central sphere point will reappear and then disappear again into the center of volume of the next frequency tetrahedron which will be a little 1 frequency tetrahedral space.

12 Green Spheres at vertices of 1 frequency cuboctahedron or Vector Equilibrium, VE:
(1,0,0) (-1,0,0) (0,1,0) (0,-1,0) (.5, -.5, .7071067) (-.5, -.5, .7071067) (.5, .5, .7071067) (-.5, .5, .707106) (.5, -.5, -.707106) (-.5, -.5, -.707106) (.5, .5, -.707106) (-.5, .5, -.707106)

6 Yellow Spheres at vertices of 2 frequency octahedron:
(1,1,0) (1,-1,0) (-1,1,0) (-1,-1,0) (0, 0, 1.414214) (0, 0, -1.414214)

16 Red Spheres at 4 frequency tetrahedron's vertices:

(1, 1.5, .707106) (1.5, -.5, .707106) (-1.5, .5, .707106) (-1.5, -.5, .707106) (1, 0, 1.41421) (2, 0, 1.41421) (-1, 0, 1.41421) (-2, 0, 1.41421) (.5, 1.5, -.707106) (-.5, 1.5, -.707106) (.5, -1.5, -.707106) (-.5, -1.5, -.707106) (0, 1, -1.41421) (0, 2, -1.41421) (0, -1, -1.41421) (0, -2, -1.41421)

1 Frequency Cube: 14 Spheres:
Add 4 Blue Spheres to the 2 frequency tetrahedron at these points to complete the 1F cube:
(0, 1, .707106) (0, -1, .707106) (1, 0, -.707106) (-1, 0, -.707106)

2 Frequency Cube: 63 Spheres:
Add 28 Spheres to the 4 frequency tetrahedron at these points to complete the 2F cube.
Add 4 Blue Spheres on each of the 4 frequency tetrahedron's faces to create ‘Star Tetrahedron’:
Add 12 Green Spheres to the Star Tetrahedron’s mid edge points at vertices of the 2 frequency cuboctahedron to complete the 2 frequency cube.

16 Blue Spheres:
(.5, 1.5, .707106) (-.5, 1.5, .707106) (.5, -1.5, .707106) (-.5, -1.5, .707106) (0, 1, 1.41421) (0, 2, 1.41421) (0, -1, 1.41421) (0, -2, 1.41421) (1.5, .5, -.707106) (1.5, -.5, -.707106) (-1.5, .5, -.707106) (-1.5, -.5, -.707106) (1, 0, -1.41421) (2, 0, -1.41421) (-1, 0, -1.41421) (-2, 0, -1.41421)

12 Green Spheres: 2 Frequency Cuboctahedron:
(2,0,0) (-2,0,0) (0,2,0) (0,-2,0) (1, 1, 1.41421) (1, -1, 1.41421) (-1, 1, 1.41421) (-1, -1, 1.41421) (1, 1, -1.414214) (1, -1, -1.41421) (-1, 1, -1.41421) (-1, -1, -1.41421)

To show all the spheres in the 2 Frequency VE or Cuboctahedron use all the 2 frequency cube’s vertices and then truncate, or delete, the following 8 sphere ‘points’ which are at the vertices of the 2 frequency cube:
(2, 0, 1.41421) (-2, 0, 1.41421) (0, 2, 1.41421) (0, -2, 1.41421) (2, 0, -1.41421) (-2, 0, -1.41421) (0, 2, -1.41421) (0, -2, -1.41421)

1 Frequency Tetrakaidecahedron: 38 Spheres:
Add 24 Orange Spheres to the 1 frequency cubes 14 spheres to complete the 1F tetrakaideca’:
(.5, .5, 1.41421) (.5, -.5, 1.41421) (-.5, .5, 1.41421) (-.5, -.5, 1.41421) (.5, .5, -1.41421) (.5, -.5, -1.41421) (-.5, .5, -1.41421) (-.5, -.5, -1.41421) (1, 1, .707106) (1, -1, .707106) (-1, 1, .707106) (-1, -1, .707106) (1, 1, -.707106) (1, -1, -.707106) (-1, 1, -.707106) (-1, -1, -.707106) (.5, 1.5, 0) (-.5, 1.5, 0) (.5, -1.5, 0) (-.5, -1.5, 0) (1.5, .5, 0) (1.5, -.5, 0) (-1.5, .5, 0) (-1.5, -.5, 0)

3 Frequency Octahedron: 44 Spheres:
Add 6 Yellow Spheres to 1 frequency tetrakaidecahedron to complete the 3F Octahedron:

(1.5, 1.5, 0) (1.5, -1.5, 0) (-1.5, 1.5, 0) (-1.5, -1.5, 0) (0, 0, 2.12132) (0, 0, -2.12132)

6 Frequency Tetrahedron: 84 Spheres:
Add 40 Blue Spheres to the 3 frequency octahedron to complete the 6F tetrahedron:
(1, 0, 2.12132) (2, 0, 2.12132) (3, 0, 2.12132) (-1, 0, 2.12132) (-2, 0, 2.12132) (-3, 0, 2.12132)
(0, 1, -2.12132) (0, 2, -2.12132) (0, 3, -2.12132) (0, -1, -2.12132) (0, -2, -2.12132)
(0, -3, -2.12132) (1.5, .5, 1.41421) (1.5, -.5, 1.41421) (2.5, .5, 1.41421) (2.5, -.5, 1.41421)
(-1.5, .5, 1.41421) (-1.5, -.5, 1.41421) (-2.5, .5, 1.41421) (-2.5, -.5, 1.41421) (.5, 1.5, -1.41421)
(-.5, 1.5, -1.41421) (.5, 2.5, -1.41421) (-.5, 2.5, -1.41421) (.5, -1.5, -1.41421) (-.5, -1.5, -1.41421)
(.5, -2.5, -1.41421) (-.5, -2.5, -1.41421) (2, 0, .707106) (2, 1, .707106) (2, -1, .707106)
(-2, 0, .707106) (-2, 1, .707106) (-2, -1, .707106) (0, 2, -.707106) (1, 2, -.707106)
(-1, 2, -.707106) (0, -2, -.707106) (1, -2, -.707106) (-1, -2, -.707106)

C) Alternative ‘Correct’ Sequencing:

This sequence of coordinates are what I consider the ‘Correct’ coordinates because this is the way the closest packed spheres within the 6 axis metric of the IVM are oriented in association with the 3 x,y,z axis. As can be seen we are linking 2 separate sets of coordinates by simply rotating the complex 45 degrees about the ‘z’ axis while also linking two separate ‘coordinate systems’, the 6 axis metric and the 3 axis metric, both of which sets of coordinates are ‘entirely arbitrary’. We could rotate the complex in any number of ways and thereby create any number of ‘entirely arbitrary’ coordinates which will all be ‘correct’ in regards to their spatial ratios within the IVM but they will be different from those listed in these 2 sections.

The difference between the 6 axis metric and the 3 axis metric is that the 6 axis metric is non-axiomatic because it is based on pre-time and pre-size vector-tensor inter-relationship patterns and therefore it does not rely on anthropic measurement associations like seconds, meters or kilograms. The numerical interrelationship patterns within the IVM are inherently Eternal and act as the underlying basis for the systematic intertransformative processes by which Nature’s geometrically structured space time continuum manifests itself to our human sense organs, as well as to our man made machines which act as extensions and additions to our mammalian sense organs.

1 Frequency Tetrahedron: 4 Spheres:
4 Red Spheres: (.353553, .353553, .353553) (-.353553, -.353553, .353553)
(.353553, -.353553, -.353553) (-.353553, .353553, -.353553)

2 Frequency Tetrahedron: 10 Spheres:
4 Red Spheres: (.7071067, .7071067, .7071067) (-.7071067, -.7071067, .7071067)
(.7071067, -.7071067, -.7071067) (-.7071067, .7071067, -.7071067)
6 Yellow Spheres = 1 Frequency Octahedron: (.7071067, 0, 0) (-.7071067, 0, 0)
(0, .7071067, 0) (0, -.7071067, 0) (0, 0, .7071067) (0, 0, -.7071067)

1 Frequency Cube: 14 Spheres:
Add 4 Blue Spheres to 2 frequency tetrahedron:
(-.7071067, .7071067, .7071067) (-.7071067, -.7071067, -.7071067)
(.7071067, .7071067, -.7071067) (.7071067, -.7071067, .7071067)

3 Frequency Tetrahedron: 20 Spheres:

4 Red Vertices of tetra: (1.06066, 1.06066, 1.06066) (-1.06066, -1.06066, 1.06066)
(1.06066, -1.06066, -1.06066) (-1.06066, 1.06066, -1.06066)

4 Blue Spheres as center = 1 frequency inverted tetrahedron:
(-.353553, -.353553, -.353553) (.353553, .353553, -.353553)
(-.353553, .353553, .353553) (.353553, -.353553, .353553)

Add 12 Purple Spheres around 4 blue spheres = 1 Frequency Truncated Tetrahedron:
(.353553, .353553, 1.06066) (-.353553, -.353553, 1.06066) (-.353553, .353553, -1.06066)
(.353553, -.353553, -1.06066) (1.06066, .353553, .353553) (.353553, 1.06066, .353553)
(-1.06066, -.353553, .353553) (-.353553, -1.06066, .353553) (1.06066, -.353553, -.353553)
(.353553, -1.06066, -.353553) (-.353553, 1.06066, -.353553) (-1.06066, .353553, -.353553)

4Frequency Tetrahedron: 35 Spheres:

1 Frequency Cuboctahedron or VE at center of 4F tetrahedron: 13 Spheres:
1 White Sphere: (0,0,0)

12 Green Spheres:
(.7071067, .7071067, 0) (.7071067, -.7071067, 0) (-.7071067, .7071067, 0)
(-.7071067, -.7071067, 0) (.7071067, 0, .7071067) (0, .7071067, .7071067)
(-.7071067, 0, .7071067) (0, -.7071067, .7071067) (.7071067, 0, -.7071067)
(0, .7071067, -.7071067) (-.7071067, 0, -.7071067) (0, -.7071067, -.7071067)

Add 6 Yellow Spheres for 19 spheres in 2 Frequency Octahedron:
(0, 0, 1.41421) (0, 0, -1.41421) (1.41421, 0, 0) (-1.41421, 0, 0)
(0, 1.41421, 0) (0, -1.41421, 0)

Add 16 Red Spheres to complete 4 frequency tetrahedron;
(1.41421, 1.41421, 1.41421) (-1.41421, -1.41421, 1.41421) (1.41421, -1.41421, -1.41421)
(-1.41421, 1.41421, -1.41421) (.707, .707, 1.414) (-.707, -.707, 1.414) (.707, -.707, -1.414)
(-.707, .707, -1.414) (1.414, .707, .707) (.707, 1.414, .707) (-1.414, -.707, .707)
(-.707, -1.414, .707) (.707, -1.414, -.707) (1.414, -.707, -.707) (-.707, 1.414, -.707)
(-1.414, .707, -.707)

2 Frequency Cube: 63 Spheres:

Add 16 blue spheres to the 4 frequency tetrahedron. Add 4 blue spheres on each of the tetrahedron's 4 faces and then add 12 green spheres at the 2 frequency cuboctahedron's vertices. These 12 spheres of the VE are also the 12 mid-edge spheres for the 2 frequency cube.

When we place the 14 points for the 1 frequency cube into this 2 frequency cube they will be situated or 'coordinated' at the centers of the 14 octahedral spaces within the 2 frequency cube. The 14 spheres of the 1F cube have 'vanished' and transformed themselves into the 14 octahedral spaces. In the 3 frequency cube these 14 octahedral spaces from the 2F cube will all transform themselves back into 14 'spheres' at the center of the 3 frequency cube. The 14 spheres of the 1F cube which 'vanished' and turned into 14 octahedral spaces will all now 'reappear' as 14 spheres at the center of the 3F cube.

This is the same thing which occurs when the 6 spheres of the 1F octahedron all 'vanish' and turn into the 6 octahedral spaces within the center of the 2F octahedron and then 'reappear' as 6 spheres at the center of the 3F octahedron.

Add 16 Blue Spheres:
(-1.41421, -1.41421, -1.41421) (1.41421, 1.41421, -1.41421) (-1.41421, 1.41421, 1.41421) (1.41421, -1.41421, 1.41421) (-.707, -.707, -1.414) (.707, .707, -1.414) (-.707, .707, 1.414) (.707, -.707, 1.414) (-1.414, -.707, -.707) (-.707, -1.414, -.707) (1.414, .707, -.707) (.707, 1.414, -.707) (-.707, 1.414, .707) (-1.414, .707, .707) (.707, -1.414, .707) (1.414, -.707, .707)

Add 12 Green Spheres at 2 Frequency Cuboctahedron's Vertices:
(1.414, 1.414, 0) (1.414, -1.414, 0) (-1.414, 1.414, 0) (-1.414, -1.414, 0) (1.414, 0, 1.414) (0, 1.414, 1.414) (-1.414, 0, 1.414) (0, -1.414, 1.414) (1.414, 0, -1.414) (0, 1.414, -1.414) (-1.414, 0, -1.414) (0, -1.414, -1.414)

After completing the 2 frequency cube we may delete the 8 vertices from its 63 total spheres to reveal the 55 spheres of the 2 frequency cuboctahedron or VE:
Delete or 'truncate' these 8 sphere points:
(1.41421, 1.41421, 1.41421) (-1.41421, -1.41421, 1.41421) (1.41421, -1.41421, -1.41421) (-1.41421, 1.41421, -1.41421) (-1.41421, -1.41421, -1.41421) (1.41421, 1.41421, -1.41421) (-1.41421, 1.41421, 1.41421) (1.41421, -1.41421, 1.41421)

1 Frequency Tetrakaidecahedron: 38 Spheres:
Add 24 Orange spheres to 1 frequency cube:
(.707, 0, 1.414) (-.707, 0, 1.414) (0, .707, 1.414) (0, -.707, 1.414) (.707, 0, -1.414) (-.707, 0, -1.414) (0, .707, -1.414) (0, -.707, -1.414) (1.414, 0, .707) (-1.414, 0, .707) (0, 1.414, .707) (0, -1.414, .707) (1.414, 0, -.707) (-1.414, 0, -.707) (0, 1.414, -.707)

(0, -1.414, -.707) (1.414, .707, 0) (.707, 1.414, 0) (1.414, -.707, 0) (.707, -1.414, 0)
(-.707, 1.414, 0) (-1.414, .707, 0) (-.707, -1.414, 0) (-1.414, -.707, 0)

12 Frequency Tetrahederon: 455 Spheres:

The 12 frequency tetrahedron is the first time that all 6 polyhedrons share the same nuclear sphere at the center of the complex, (0,0,0). The 12F tetrahedron has the 1 frequency VE at the center. This has a 2 frequency cube and a 2 frequency VE around it. This 2F cube has a 3F VE around it and this has a 2 frequency tetrakaidecahedron around it. The 2F tetrakaidecahedron has a 6 frequency octahedron around it and this has a 4 frequency truncated tetrahedron around it. I will list some of these coordinates so we can generate this omni-concentric hierarchy of inter-phasing polyhedrons as a computer generated image.

Important note:

You may have noticed that all coordinates related to the vertices of all frequencies of these 6 symmetrical polyhedrons are always whole rational increments of the first coordinates.

With this in mind it is very easy to calculate these 6 polyhedron's continuing frequencies 'prime vertices'. The prime vertices are the total number of spheres as vertices in the first frequency of the originating polyhedrons.

The tetrahedron has 4 prime vertices, the octahedron has 6, the truncated tetrahedron has 12, the cube has 8, The cuboctahedron, or VE, has 12, and the tetrakaidecahedron has 24.

Simply multiply any of these polyhedrons prime vertices (x,y,z) Cartesian coordinates by 2, 3, 4, 5, 6, 7 etc. to obtain all of the continuing frequencies prime vertexial x,y,z coordinates for these polyhedrons in association with how they are oriented within the IVM of closest packed unit radius spheres.

For example the first 4 vertices of the 1 frequency tetrahedron are:
(.353553, .353553, .353553) (-.353553, -.353553, .353553) (.353553, -.353553, -.353553)
(-.353553, .353553, -.353553)

The 2 frequency tetrahedron's 4 prime vertices are at:
(.7071067, .7071067, .7071067) (-.7071067, -.7071067, .7071067)
(.7071067, -.7071067, -.7071067) (-.7071067, .7071067, -.7071067)

The 3 frequency tetrahedron's 4 prime vertices are at:
(1.06066, 1.06066, 1.06066) (-1.06066, -1.06066, 1.06066)
(1.06066, -1.06066, -1.06066) (-1.06066, 1.06066, -1.06066)

The 4 frequency tetrahedron's 4 prime vertices are at:

(1.41421, 1.41421, 1.41421) (-1.41421, -1.41421, 1.41421) (1.41421, -1.41421, -1.41421)
(-1.41421, 1.41421, -1.41421)

As each sphere within the IVM acts as an external or internal vertex of some polyhedral shape we can understand that the ‘illions’ of spheres which accumulate within these polyhedrons will all have exact coordinates within the 3 axis metric.

By mapping out the external domain of these polyhedrons we can gain a visually apprehensible geometrical model of Fuller’s omni-concentric hierarchy of omni-directionally expanding polyhedrons.

By plotting their x,y,z coordinates into a computer program we will help new students to understand how the whole rational accounting methodology based on synergetics vectors, tensors and volumes are realized within the ‘structure’ of the IVM’s omni-concentric hierarchy of rationally phasing polyhedral continuums as they all develop as prime elements of Universal structure.

Use the 2 Frequency Cube as the center of the 12 frequency tetrahedron and place the following points around this cube. This cube’s 8 vertices are at the center of the 6 frequency octahedron’s 8 faces. They are also at the center of the 1 frequency tetrakaidecahedron’s 8 hexagonal faces. The 4 blue vertices of this 2 frequency cube are the 4 blue vertices of the inverted 4 frequency tetrahedron and they are located at the center of the 12 frequency tetrahedron’s 4 faces.

12 Frequency Tetrahedron’s 4 vertices in Red:
(4.242640687, 4.242640687, 4.242640687) (-4.242640687, -4.242640687, 4.242640687)
(4.242640687, -4.242640687, -4.242640687) (-4.242640687, 4.242640687, -4.242640687)

4 Frequency Truncated Tetrahedron’s 12 Vertices in Purple:
(1.414, 1.414, 4.242640687) (-1.414, -1.414, 4.242) (-1.414, 1.414, -4.242)
(1.414, -1.414, -4.242) (4.242, 1.414, 1.414) (1.414, 4.242, 1.414)
(-4.242, -1.414, 1.414) (-1.414, -4.242, 1.414) (4.242, -1.414, -1.414)
(1.414, -4.242, -1.414) (-1.414, 4.242, -1.414) (-4.242, 1.414, -1.414)

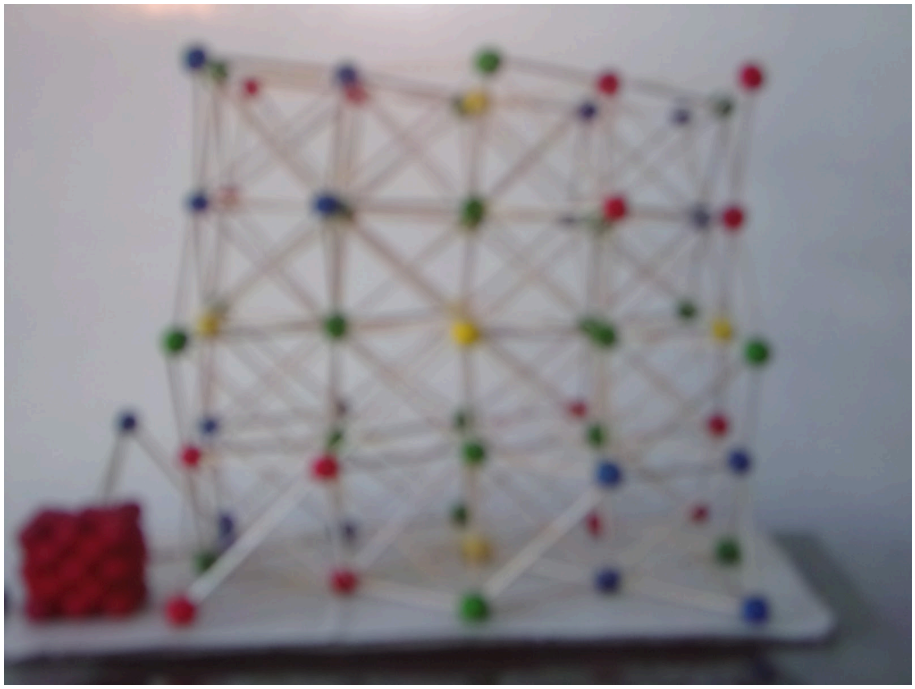
6 Frequency Octahedron’s 6 Vertices in Yellow:
(4.242, 0, 0) (-4.242, 0, 0) (0, 4.242, 0)
(0, -4.242, 0) (0, 0, 4.242) (0, 0, -4.242)

3 Frequency Cuboctahedron’s 12 Vertices in Green:
(2.121320344, 2.121320344, 0) (2.121320344, -2.121320344, 0)
(-2.121320344, 2.121320344, 0) (-2.121320344, -2.121320344, 0)
(2.121320344, 0, 2.121320344) (0, 2.121320344, 2.121320344)
(-2.121320344, 0, 2.121320344) (0, -2.121320344, 2.121320344)

(2.121320344, 0, -2.121320344) (0, 2.121320344, -2.121320344)
(-2.121320344, 0, -2.121320344) (0, -2.121320344, -2.121320344)

2 Frequency Tetrakaidecahedron: 24 Orange Sphere Vertices:
(1.414213562, 0, 2.828427125) (-1.414, 0, 2.828) (0, 1.414, 2.828) (0, -1.414, 2.828)
(1.414, 0, -2.828) (-1.414, 0, -2.828) (0, 1.414, -2.828) (0, -1.414, -2.828) (2.828, 0, 1.414)
(-2.828, 0, 1.414) (0, 2.828, 1.414) (0, -2.828, 1.414) (2.828, 0, -1.414) (-2.828, 0, -1.414)
(0, 2.828, -1.414)(0, -2.828, -1.414) (2.828, 1.414, 0) (1.414, 2.828, 0) (2.828, -1.414, 0)
(1.414, -2.828, 0)(-1.414, 2.828, 0) (-2.828, 1.414, 0) (-1.414, -2.828, 0) (-2.828, -1.414, 0)

Figure 3: A 2 Frequency Cube: 63 tensor spheres, 240 vectors, 192 tetravolume.



The picture above shows the 2 frequency cube with the same colored balls as in the coordinates listed above. The computer models should all look like the polyhedrons I have displayed.

Figure 4 below shows the 6 frequency tetrahedron. It has the 3 frequency octahedron's 6 yellow vertices at the mid edge length and the 1F tetrakaidecahedron is shown with 24 green spheres.

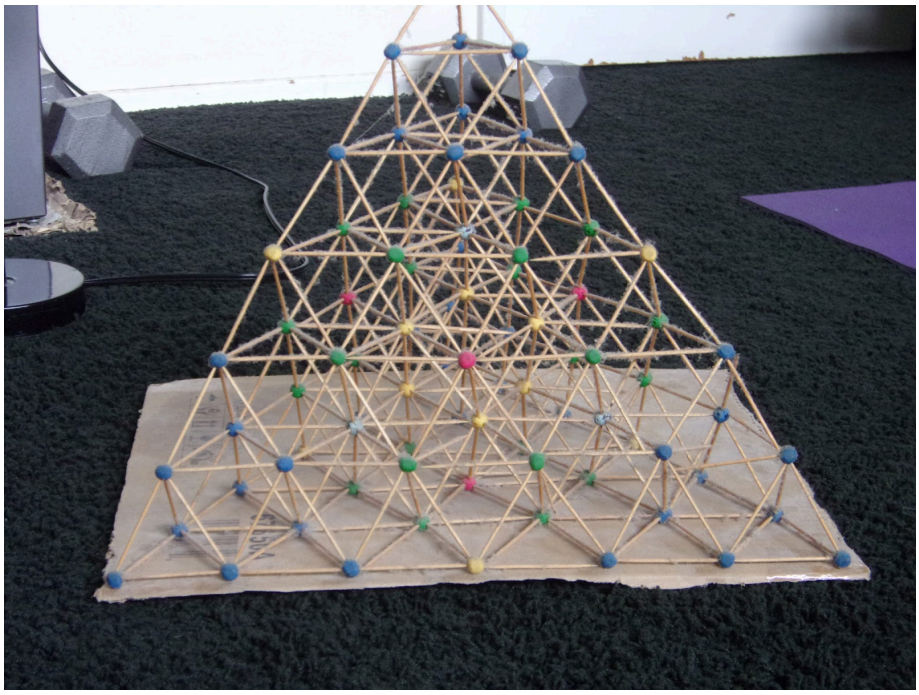
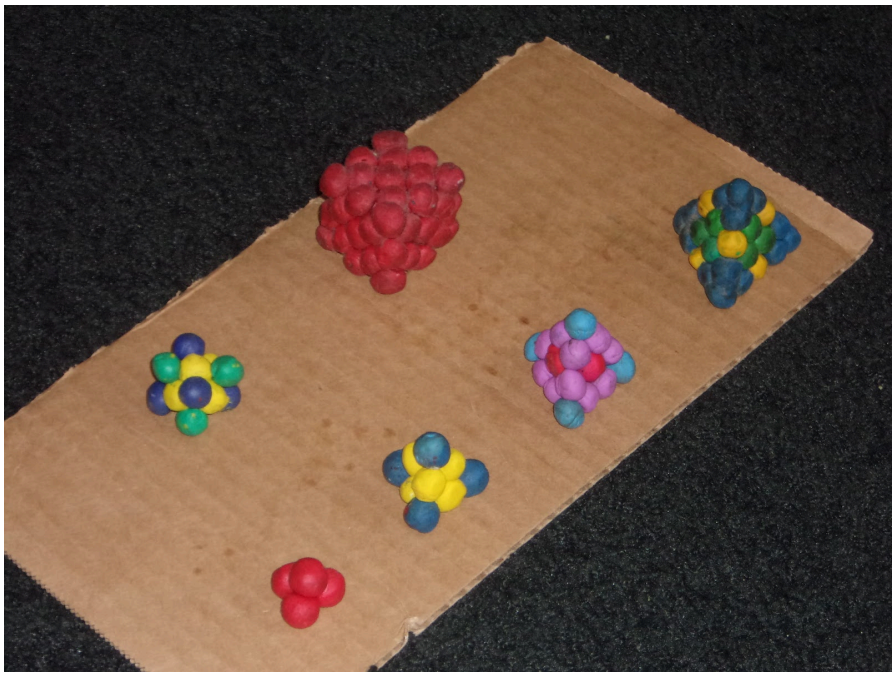


Figure 5: The first four sizes or frequency of tetrahedron and the first two frequency cubes.



The coordinates are listed so the cube will sit on its face and the other polyhedrons will be sitting up on their vertices or edges or faces. The octahedron is up on a vertex. The tetrahedron and truncated tetrahedron are on their edges. The tetrakaidecahedron and cuboctahedron sit upon a square face.

Figure 6: 1F cube, 2F cube, 4F tetrahedron, 1F tetrahedron.



D) Orienting the 2 Separate sets of xyz Coordinates:

The picture of the 6 frequency tetrahedron in Figure 7 below helps us to see the 2 different ways I am orienting the tetrahedron to the xyz coordinate system.

The “First Orientation” is set by associating the x and y axis as the tetrahedron’s 2 edge lengths which we see running from left to right and bottom to top of the tetrahedron pictured below in Figure 7.

The yellow ball at the center is the 3 frequency octahedron's ‘top’ vertex on the z axis at (0, 0, 2.12132). From top to bottom and from left to right we can count the 7 spheres and 6 vectors running along the x and y axis of the 6 frequency tetrahedron at 1 inch intervals. This is why I first placed the coordinates on the tetrahedron’s edge lengths. It makes it easier to orient ourselves with the 1 unit vector, or the 1 inch edge length of the 1 frequency tetrahedron.

The 4 yellow spheres form a square at the center of the 3F octahedron and are at: (1.5, 1.5, 0) (1.5, -1.5, 0) (-1.5, 1.5, 0) (-1.5, -1.5, 0) The 3F octahedron’s 6th hidden yellow vertex is at the bottom of the complex at (0, 0, -2.12132).

The “Second Orientation” is “Correct” and is set by associating the x and y axis as the diagonals which run between the octahedron’s central square plane. There are 4 yellow spheres at the center of the tetrahedron’s 4 external edge lengths. These 4 yellow spheres form the square at the center of volume of the octahedron. Connecting the oppositely oriented vertices gives us the 2 xy axis running at 90 degrees to one another. The z axis runs from the central yellow ‘top’ sphere point to the bottom yellow point which is hidden behind the top yellow sphere.

Notice there are 3 vector intervals between the 4 yellow spheres which form the central square plane within the 3 frequency octahedron. In this orientation we are looking ‘down’ from the top of the z axis and this helps us to orient ourselves as to the positioning of the IVM spheres of the 6 axis metric within the 3 x,y,z axis system.

Figure 7: 6F Tetrahedron.

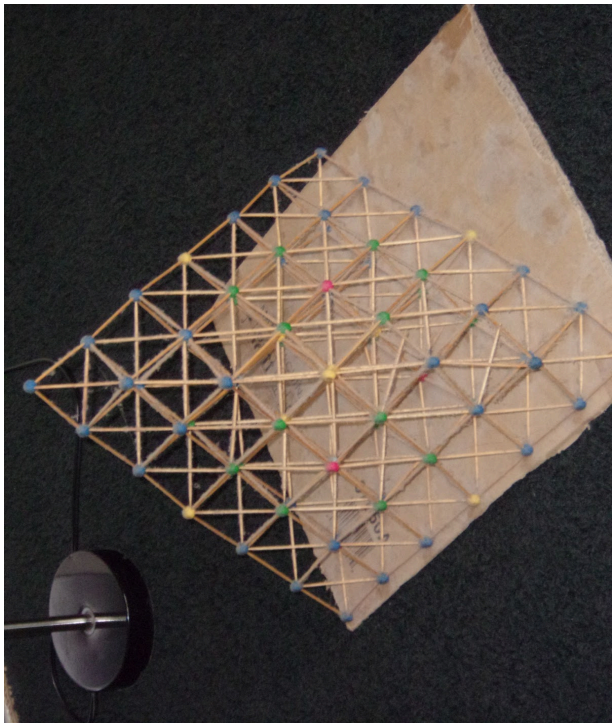
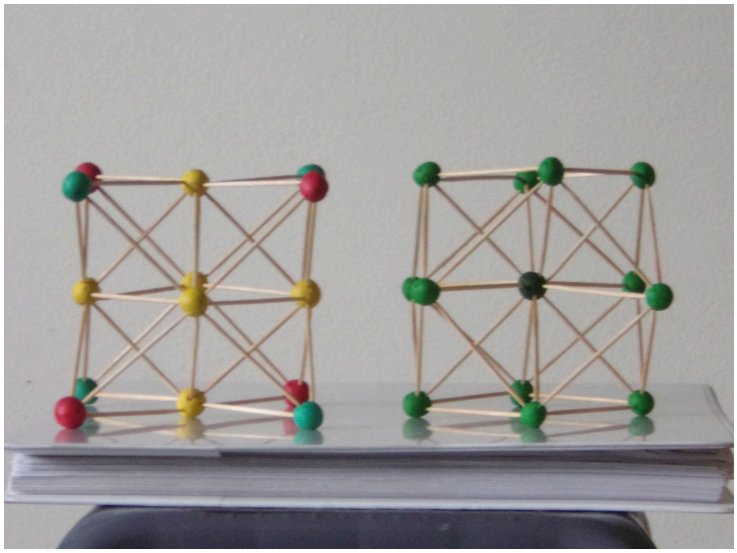


Figure 1: 1 frequency cube and 1 frequency cuboctahedron or VE.



There are 36 vectors interconnecting the 14 spheres of the cube on the left and there are also 36 vectors interconnecting the 13 spheres of the VE on the right. If a computer programmer can rotate all 36 vectors 90 degrees they will be able to show the transformation of the cube into the VE as a relation to the transformation of a ‘spin angular momentum vector’ from quantum mechanics which will show how the 14 spheres of the cube can transform themselves into the 14 faces of the VE. When we spin the 36 vectors of the newly formed VE another 90 degrees

the 13 spheres of the VE all transform themselves into the center of the 13 octahedral spaces associated with the cube. There is an obvious octahedral space at the center of the cube on the left. The 12 mid edge length ‘points’ of the cube are also at the center of volume of an octahedron but the full octahedral space is not visible in this isolated image of the cube’s edge as one ‘quantum unit’ of ‘cubed space’.

If a programmer only rotates the 6 edge lengths associated with the 2 frequency tetrahedron’s edges by 90 degrees this will transform the red tetrahedron into the oppositely oriented blue tetrahedron. The first four red vertex spheres will disappear and the vectors will rotate about the yellow point at the center of the cube and find their next ‘hold and stay station’ at the four blue ‘tensor points’. At relativistic speeds this transformational spinning vectorial design will reveal the internally patterned cube as the tetrahedron’s vector edge lengths spin into and out of the vertices of the oppositely oriented tetrahedron. These vertices to the polyhedral system act as momentary hold and stay stations where the spinning energy vector units momentarily display themselves as self-stabilizing inter-triangulated pattern-integrity polyhedral structures.

What we come to understand from these intertransformative models is that Universe has a way to create its quantum polyhedral units of structure as continuously phasing vector spins to first stabilize themselves at one polyhedral point complex and then another. These polyhedral systems are continuously phasing into ‘discrete’, individual and ‘discontinuous’ polyhedral point complexes at relativistic speeds and so the position and momentum of these points becomes difficult to calculate and thereby confuses the human mind into its ‘uncertainty paradox’.

Universe uses this ‘design stratagem’ to create time and space by continuously reshuffling the discontinuous ‘point particles’ of the polyhedral systems one into another by spinning its quantum unit vectors and ‘jump phase transitioning’ these point particles into wave pattern integrities until they ‘reorient’ themselves at the next stable hold and stay station, which in this example certainly and deterministically transforms the cubes vertices into the spaces of the cuboctahedron and vice versa, thereby enabling two stable polyhedral systems the ability to jump phase transition their energy momentum vectors, which are traveling at relativistic speeds, into self-stabilizing inter-triangulated polyhedral wave pattern integrities, first stabilizing themselves moment’um’arily as a cube and then as a cuboctahedron. It is at first ‘certainly’ a cube and then it ‘determines’ to spin its vectors 90 degrees so it can ‘momentarily’ transform itself into a ‘certainly determined’ cuboctahedron, until it retransforms itself into the cube again. The momentarily stable polyhedrons are all moment’um’arily vanishing and reappearing over and over again, certainly and deterministically, or else we wouldn’t be here!

“Mapping the Vanishing and Reappearing Quanta Points of the IVM’s 6 Axis Metric Upon the Cartesian Coordinate System’s 3 x,y,z axis”
All Rights Reserved by Noel Coughlin
© Copyright September, 2025